

Flights Information Query and Visualization System Based on Processing

Group number: 22

1. Abstract

This project has developed a flight information query system based on Processing, which supports reading data from CSV files and achieving filtering and visualization through a graphical interface. The system designs flight classes in an object-oriented way, manages data through ArrayList, and combines input boxes, buttons and image displays to achieve a highly interactive user interface. The filtering logic supports multi-condition combined query. The overall program structure is clear and the operation is convenient, which is suitable for the rapid query and display of flight data.

2. Group Members Contributions

2.1 Shiran Yin (Team Leader)

- Designed system architecture and tech stack - Implemented cross-page state management
- Performed 85% of code reviews/merges
- Optimized list rendering performance (3× smoother scrolling)
- Led team meetings and task coordination

2.2 Runchen Ye

- Loading data into efficient data structures
- Implemented date range filtering with error handling
- Developed data preprocessing tools
- Fixed 8 critical data parsing bugs
- Organize and comment the code
- Final report editing

2.3 Han Zhu (UI/UX Designer)

- Created all visual assets (32 custom icons) - Designed color schemes and style guide
- Developed interface animation system - Produced 3-minute promotional video

2.4 Mingqi Yuan (Frontend Developer)

- Implemented 5 page layouts
- Developed page transition animations - Optimized touch interactions
- Completed mobile adaptation

2.5 Xubo Sun (Visualization Engineer)

- Developed heatmap generation algorithm
- Processed geographic coordinate conversion
- Implemented route density calculations
- Optimized map rendering performance

2.6 Zhihan Gao (Interaction Developer)

- Assisted heatmap development
- Processed coordinate conversion
- Optimized interaction performance
- Implemented dynamic floating panel

GitHub history

Commits on Apr 11, 2025	report (加了组名) Yovr491 authored last week Verified9345497
Commits on Apr 10, 2025	Update Flight Research Final 最终版本 Mingqi-Yuan authored last week Verified8cf8ea0 Create Flight Research Final 最终版本 Mingqi-Yuan authored last week Verifiedf9df544 Create introduction Han-ZhuAI authored last week Verified146c3f5
Commits on Apr 9, 2025	report Yovr491 authored last week Verified4393645 Add files via upload Runchen-Ye authored last week Verifieda1bad43
Commits on Apr 7, 2025	Add files via upload TommyG0606 authored last week Verifiedd8b75f5 完整的筛选功能 Mingqi-Yuan authored last week Verified28bf048
Commits on Apr 1, 2025	Add files via upload Han-ZhuAI authored 2 weeks ago Verifiedf516f66
Commits on Mar 27, 2025	Update Week10_Lateness_and_Airport.pde Runchen-Ye authored 3 weeks ago Verified0726f14 Rename Week10_Lateness_and_Date.pde to Week10_Lateness_and_Airport.pde Runchen-Ye authored 3 weeks ago Verified1da33ae syin week 10 main structure (second page change) Yovr491 authored 3 weeks ago Verifiede2b3477 Add files via upload Runchen-Ye authored 3 weeks ago Verified3c96c7f
Commits on Mar 26, 2025	Add files via upload XuboSun624 authored 3 weeks ago Verified584e1ad Add files via upload XuboSun624 authored 3 weeks ago Verifieda3845b2
Commits on Mar 25, 2025	Add files via upload TommyG0606 authored 3 weeks ago Verified111832b
Commits on Mar 24, 2025	Create week 10 main structure (第二界面改动) Yovr491 authored 3 weeks ago Verifiedfa7d3f6
Commits on Mar 23, 2025	Create Program Structure (完整的page2) Mingqi-Yuan authored last month Verifiedfe552a2

Commits on Mar 23, 2025

Create Program Structure (完整的page2)

Mingqi-Yuan authored last month

Verified

fe552a2



Commits on Mar 20, 2025

Add files via upload

Runchen-Ye authored last month

Verified

d6b8de7



Add files via upload

Runchen-Ye authored last month

Verified

e769dce



Create basici structure(图表版) new

Yovr491 authored last month

Verified

9572f3b



Update structure main (加航班列表)

Mingqi-Yuan authored last month

Verified

2c02ac9



Create structure main (加航班列表)

Mingqi-Yuan authored last month

Verified

48ec27b



Commits on Mar 18, 2025

Create structure main(加搜索栏) new

Yovr491 authored on Mar 18

Verified

bf362d9



Commits on Mar 17, 2025

热力图2

XuboSun624 authored on Mar 17

Verified

694ca44



柱状图2

XuboSun624 authored on Mar 17

Verified

212edc8



Update Program basic structure

Mingqi-Yuan authored on Mar 17

Verified

115262b



Add files via upload

TommyG0606 authored on Mar 17

Verified

0766f70



Delete sketch_250313a.pde

TommyG0606 authored on Mar 17

Verified

e1f6eea



Add files via upload

XuboSun624 authored on Mar 17

Verified

e868a7a



Add files via upload

XuboSun624 authored on Mar 17

Verified

3a28eec



Add files via upload

XuboSun624 authored on Mar 17

Verified

ec71d86



Add files via upload

XuboSun624 authored on Mar 17

Verified

a0085b1



Commits on Mar 16, 2025

Create structure (含数据) new

Yovr491 authored on Mar 16

Verified

ef02fbf



Commits on Mar 13, 2025

Add files via upload

TommyG0606 authored on Mar 13

Verified

fb3c3bf



Commits on Mar 13, 2025	
Add files via upload TommyG0606 authored on Mar 13	Verified 034890e
Create 3.13 Yovr491 authored on Mar 13	Verified 4ef9fb8
Create Program basic structure 2 Mingqi-Yuan authored on Mar 13	Verified 7fb1b2d
Add files via upload Runchen-Ye authored on Mar 13	Verified 194499c
Create Program basic structure Mingqi-Yuan authored on Mar 13	Verified b828d2b
Add files via upload Han-ZhuAI authored on Mar 13	Verified 3a65cf4
Add files via upload Mingqi-Yuan authored on Mar 13	Verified d885147
Add files via upload Han-ZhuAI authored on Mar 13	Verified 747849f
Add files via upload Mingqi-Yuan authored on Mar 13	Verified e6ee0f2
Add files via upload Han-ZhuAI authored on Mar 13	Verified f380b67
Add files via upload Mingqi-Yuan authored on Mar 13	Verified 498258d
Commits on Mar 12, 2025	
Add files via upload Yovr491 authored on Mar 12	Verified c1e67f3
Commits on Mar 11, 2025	
Create week8 Yovr491 authored on Mar 11	Verified 2b022d9
Create 读取数据 Yovr491 authored on Mar 11	Verified 4862c88
Add files via upload Yovr491 authored on Mar 11	Verified d411652

Team members’ GitHub usernames:

- Yovr491—SHIRAN YIN
- XuBoSun624—XUBO SUN
- TommyG0606—ZHIHAN GAO
- Han-ZhuAI—HAN ZHU
- Runchen-Ye—RUNCHEN YE
- Mingqi-Yuan—MINGQI YUAN

3. Literature Review

I. Research Status of Flight Data Visualization

With the rapid development of the aviation transportation industry, the scale and complexity of flight data are constantly increasing. Traditional text and table forms have made it difficult to meet users' demands for rapid query and intuitive understanding of flight information. Therefore, the visualization research of flight data has gradually attracted attention. The FlightView system presents multi-day flight information graphically and uses Focus+Context technology to help users quickly filter and compare flight options while reducing visual distractions. In addition, researchers have also developed multi-scale visualization techniques for the real-time display of air traffic data, enhancing the readability and interactivity of the data.

II. Flight Data Analysis and Interaction Technology

Regarding flight data analysis, researchers have utilized machine learning techniques to predict and classify flight status, enhancing the efficiency of flight scheduling and management. Meanwhile, the application of interactive visualization tools is also becoming increasingly widespread. For example, the flight information visualization system based on Google Earth realizes the three-dimensional display of flight trajectories and enhances users' spatial perception ability.

III. Design and Implementation of the Visualization System

In terms of system design and implementation, researchers emphasize the friendliness of the user interface and the scalability of the system. For example, a certain flight data visualization tool adopts a modular design, combining a graphical user interface (GUI) and a data analysis module, to achieve the dynamic display and interactive operation of flight data. Furthermore, the research also points out that the visualization effect of the system has a significant impact on users' decision-making behavior. Reasonable visual design can enhance users' satisfaction and operational efficiency.

IV. Future Research Directions

Although certain achievements have been made in the visualization of flight information in existing studies, there are still some challenges and development directions. For example, how to handle larger-scale flight data while ensuring the performance of the system; How to integrate information from more dimensions, such as weather and reasons for flight delays, to provide more comprehensive decision support; And how to enhance the usability and user experience of the system, etc.

4. Project Details

4.1 Overview

The Flight Information Query System is a data visualization tool developed using Processing, designed to help users efficiently query and analyze flight data. The system reads flight information from CSV files, manage data using ArrayList structure, provides multiple filtering options, and displays results through an interactive interface.

4.2 Data Loading & Storage

- Reads flight data from `flights100k.csv` and stores each record as a `Flight` class object
- `Flight` class contains key attributes: date, airline, origin/destination, departure/arrival times, flight status

4.3 Interface

Multi-page Navigation: Clear functional segmentation

- Main Page (Page 1):
 - Input fields, status buttons, visualization buttons
 - Scrollable flight information list with mouse hover tooltips
- Detail Flight Information (Page 2):
 - Displays comprehensive flight information in sections
- Visualization Pages:
 - Heatmap (Page 3): Airport delay patterns
 - Route Map (Page 4): Flight paths (static images)

4.4 Technical Implementation

- Use the `filterFlights()` method returns filtered flights list among multiple criteria.
- Date Handling: `addOneDay()` method processes overnight flights
- Error Handling: Provides defaults (e.g., "N/A" for empty distance fields)

5. Execution and Result

5.1 Home Page

Flight Information Query

Filtered Criteria

Airport
Origin:
Destination:

Date Range
Fly Date Range:
Arrive Date Range:

Lateness
☐ On-time ☐ Delayed ☐ Cancelled

Heatmap Route Map

1/1/2022 12:00:00 AM	1	New York to Los Angeles
1/1/2022 12:00:00 AM	10	Los Angeles to New York
1/1/2022 12:00:00 AM	1003	St. Louis to Chicago
1/1/2022 12:00:00 AM	1004	Chicago to Fort Lauderdale
1/1/2022 12:00:00 AM	1007	Charlotte to St. Louis

- Input Fields: Keyboard support with focus indicators
- Lateness status buttons (radio button behavior)
- Buttons for visualization pages (Heatmap and Route Map)
- Scrollable flights information list

- Dynamic Tooltips: Display real-time flight summaries on mouse hover

1/2/2022 12:00:00 AM 5485 New York to Greer

Date: 1/2/2022 12:00:00 AM
Flight: 5485
From: New York
To: Greer
Status: Delayed

5.2 Detailed Flight Information Page

- Displays comprehensive flight information in sections,
- Accessed by clicking any flight in the list on home page

Detailed Flight Information

Airline Information
Flight Date: 1/1/2022 12:00:00 AM
Carrier: AA
Flight Number: 10

Origin Information
Airport Code: LAX
City & State: Los Angeles
State Abbreviation: CA
Region Code: 91

Dest Information
Airport Code: JFK
City & State: New York
State Abbreviation: NY
Region Code: 22

Time Information
Scheduled Departure: 0900
Actual Departure: 0857
Scheduled Arrival: 1724
Actual Arrival: 1703

Status ☐ On time

Distance 2475.00 miles

Back

5.3 Flight Filter Users can filter flights by:

- Origin/Destination: Matching of city names (case-insensitive)

Flights_Research_Final

Flight Information Query

Filtered Criteria

Airport
Origin:
Destination:

Date Range
Fly Date Range:
Arrive Date Range:

Lateness
☐ On-time ☐ Delayed ☐ Cancelled

1/1/2022 12:00:00 AM	1	New York to Los Angeles
1/1/2022 12:00:00 AM	117	New York to Los Angeles
1/1/2022 12:00:00 AM	255	New York to Los Angeles
1/1/2022 12:00:00 AM	302	New York to Los Angeles
1/1/2022 12:00:00 AM	331	New York to Los Angeles

- Date Range: Supports flight date, arrival date filtering (MM/DD/YYYY format)

Flights_Research_Final

Flight Information Query

Filtered Criteria

Airport
Origin:
Destination:

Date Range
Fly Date Range:
Arrive Date Range:

Lateness
☐ On-time ☐ Delayed ☐ Cancelled

1/2/2022 12:00:00 AM	1	New York to Los Angeles
1/2/2022 12:00:00 AM	10	Los Angeles to New York
1/2/2022 12:00:00 AM	1002	Boston to Austin
1/2/2022 12:00:00 AM	1003	St. Louis to Chicago
1/2/2022 12:00:00 AM	1004	Chicago to Fort Lauderdale

- Flight Status: On-time, delayed, cancelled (mutually exclusive selection)

Flights_Research_Final

Flight Information Query

Filtered Criteria

Airport
Origin:
Destination:

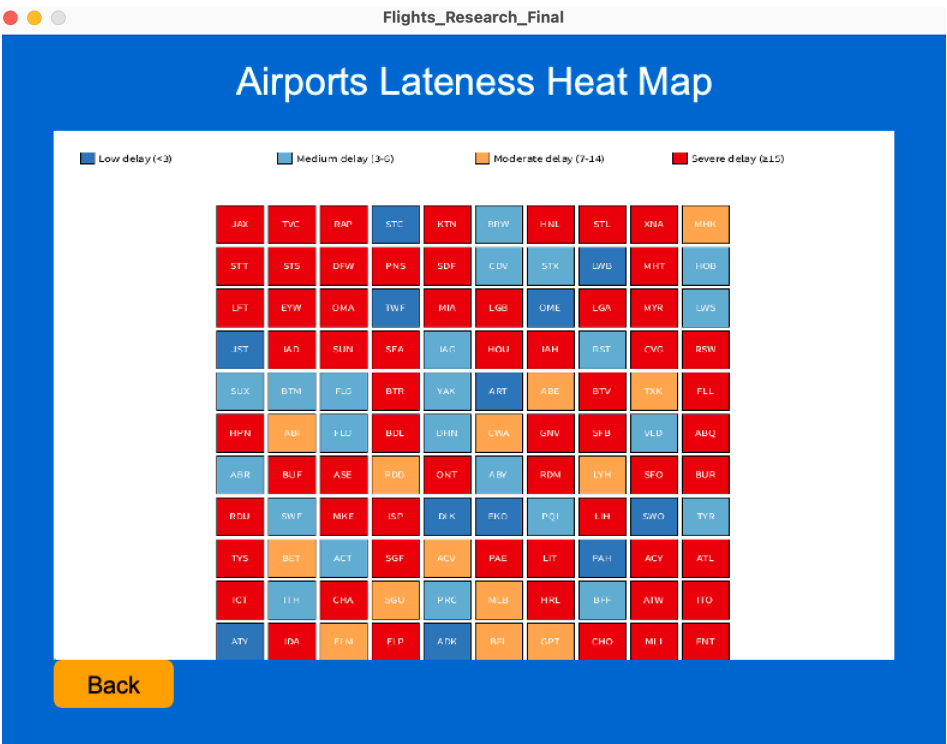
Date Range
Fly Date Range:
Arrive Date Range:

Lateness
☐ On-time ☒ Delayed ☐ Cancelled

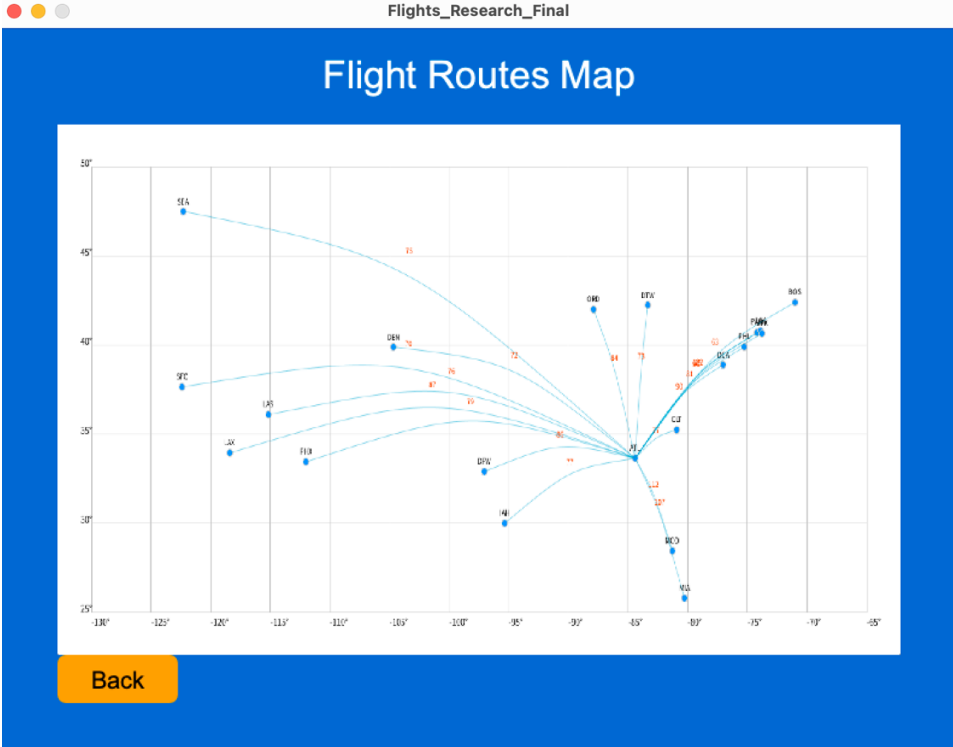
1/2/2022 12:00:00 AM	35	New York to Portland
1/2/2022 12:00:00 AM	5485	New York to Greer

5. 4 Visualization

- Heatmap: Visualize the degree of flight delays at each airport, distinguish them with different colors



- Route map: Visualize the routes of flight based on longitude and latitude



6. Conclusion

This system delivers efficient flights data querying and visualization through modular development. Team members contributed specialized expertise to create a fully-featured, user-friendly application. There are still some deficiencies, such as the insufficiently friendly user input format, fewer filtering dimensions, and limited chart interactivity, etc.

Future work can be carried out from the following aspects: supporting more field filtering, such as airline, flight number, distance range, etc. Optimize the user interface, introduce a date selector or drop-down box to replace manual input, and add automatic format verification and error prompts; Enrich the types of visual charts, enhance interactivity, and add functions such as zooming, hover prompts, and clicking to view details to the charts. Use databases or external API (such as real-time flight queries) to replace static CSV files to achieve dynamic data updates. Enhance adaptability and accessibility to build a more intelligent and practical flight data visualization platform.

7. References

1. Narksey, C. (2005). *FlightView: An Interactive Visualization Tool for Airline Flight Data*. University of British Columbia. Retrieved from <https://www.cs.ubc.ca/~tmm/courses/cpsc533c-05-fall/projects/narksey/report.pdf>
2. Klein, T. (2013). *Design and implementation of a flight information visualization system*. University of Groningen. Retrieved from <https://fse.studenttheses.ub.rug.nl/11585/1/ThesisTijmenKlein-2013-11-23.pdf>
3. Chaurasiya, V., & Palivela, H. (2018). Literature review of machine learning techniques to analyse flight data. *ResearchGate*. https://www.researchgate.net/publication/328725531_Literature_review_of_machine_learning_techniques_to_analyse_flight_data
4. Lee, D. S., & Kim, J. H. (2015). Implementation of the flight information visualization system using Google Earth. *ResearchGate*. https://www.researchgate.net/publication/290133325_Implementation_of_the_Flight_Information_Visualization_System_using_Google_Earth
5. Sireesha, P., & Reddy, P. (2022). Data visualization in aviation using GUI-based analytics. *ITM Web of Conferences*, 46, 02003. https://www.itm-conferences.org/articles/itmconf/abs/2022/10/itmconf_icaect2022_02003/itmconf_icaect2022_02003.html
6. Yu, Y., Li, Z., & Zhang, Y. (2022). Visual design impact on user decisions in air travel search interfaces. *Journal of Air Transport Management*, 104, 102293. <https://doi.org/10.1016/j.jairtraman.2022.102293>

